

Joseph Khalaf

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EDUCATION

Michigan State University – E. Lansing, MI

GPA: 3.82

Bachelor of Science, Computer Science | Minor in Business

May 2025

EXPERIENCE

Little Caesars – Detroit, MI

Backend Software Engineering Intern | Marketing Technologies

May 2025 – Current

- Created service level objectives (SLOs) using Nobl9 and Grafana to monitor reliability and performance of the fundraising website by logging custom metrics throughout the existing, large codebase.
- Contributed to CI/CD pipelines using Azure DevOps and YAML, streamlining deployments and improving release reliability.
- Wrote extensive unit tests to improve test coverage for the repository and prevent regression issues.

General Motors – E. Lansing, MI

Computer Science Capstone

January 2025 – May 2025

- Developed a production-ready web dashboard via React, Python, and Flask that used AI and data analysis to detect anomalies in GM's waste management system for their manufacturing plants.
- Led team workflow by setting weekly goals, reviewing code, and acting as the main liaison with GM, including leading client meetings and the final project presentation.
- Created documentation to support a smooth handoff to GM and long-term maintainability.

PROJECTS

SpartaHack 8 – E. Lansing, MI

Jan 2023

- Programmed an AI using TensorFlow to detect the location of potholes based on an image of a road.

MHacks 16 – Ann Arbor, MI

Nov 2023

- Developed a real-time speech-to-text translation system overlaying subtitles below the speaker using computer vision and augmented reality to simplify communication with other languages.

Connect-4 AI | 1st Place Project – E. Lansing, MI

Sep 2024 – Dec 2024

- Designed and trained AI models to play Connect-4 using Reinforcement Learning, CNN, MLP, SVM, Decision Tree, and Minimax using various python libraries such as scikit-learn, PyTorch, and TensorFlow.

Robofest | 3rd Place at Worlds – Troy, MI

Sep 2018 – May 2022

- Designed and coded robots in C to compete in autonomous games while planning team building events, managing team projects, and teaching other students the basics of coding in C.

LEADERSHIP & CLUB ACTIVITY

- Robofest Programming Lead, Robotics Club – Troy Athens High School
- Member, AI Club – Michigan State University
- Member, SJP MSU – Michigan State University

Sep 2019 - May 2022

Sep 2022 - May 2025

Sep 2024 - May 2025

SKILLS

Programming Languages: Python, C++, C#, C, JavaScript, TypeScript, Java, ARM Assembly, SQL, HTML, CSS

Frameworks & Tools: React, PyTorch, TensorFlow, scikit-learn, Flask, Docker, YAML, Git, Unix, Android Studio, MongoDB, MySQL, Grafana, Nobl9, CI/CD Pipelines, Azure DevOps